MOHAMED MUFASSIR

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PROFESSIONAL SUMMARY

3rd-year Computer Systems Engineering undergraduate (SLIIT) with a strong foundation in embedded systems, real-time control, and robotics. A fast learner who picks up new tools and frameworks quickly — from AVR assembly to ROS 2 — and applies them to build working hardware-software systems, not just theoretical exercises. Combines hands-on project experience with a genuine drive to keep expanding into new domains of robotics and automation.

EDUCATION

BSc (Hons) in Computer Systems Engineering Nov 2023 – Expected Nov 2027
Sri Lanka Institute of Information Technology (SLIIT)

Relevant Modules: Real-Time Systems, Digital Logic Design, Control Theory, Computer Architecture, Data Structures & Algorithms, Computer Networks

EXPERIENCE

Software Engineering Intern — ZILLIT (Startup), Nittambuwa *Nov 2025 – Feb 2026*

- Led sole development of a cross-platform Point of Sale (POS) desktop application using Electron.js and TypeScript, designed as part of a broader IoT ecosystem (desktop + mobile integration)
- Built the front-end of an enterprise-level financial management system using Vue.js, integrating with a Java Spring Boot backend
- Worked with React, TypeScript, Vue.js, Electron.js, PostgreSQL, and Docker in a fast-paced startup environment
- Collaborated on system architecture decisions for scalable, production-ready software

TECHNICAL SKILLS

- **Embedded & Hardware:** AVR (ATmega), Arduino, ESP32, Raspberry Pi, UART / SPI / I2C, Sensors & Actuators, PCB Design, Digital Logic Design, Oscilloscope & Lab Instruments
- **Languages:** C/C++, Embedded C, AVR Assembly, Python, JavaScript, TypeScript, Rust (Basic)
- **Robotics & Simulation:** ROS 2 (in progress), Fusion 360, Proteus, 3D CAD Design
- **Tools:** Git, Linux OS, Arduino IDE, MATLAB, VS Code, EasyEDA
- **Software Stack:** React, Vue.js, Node.js/Express.js, Spring Boot (Java), PostgreSQL/MySQL, Docker (basic)

PROJECTS

PID-Based DC Motor Speed Control System (Raspberry Pi + Python) *2026*

- Designed a real-time closed-loop control system using PID algorithm with encoder feedback for precise RPM measurement and regulation
- Extended the project with a Python desktop GUI application to visualize live motor behavior, RPM graphs, and PID tuning parameters in real time
- Achieved stable, accurate motor control — successfully demonstrated and documented system performance
- Stack: Python, Raspberry Pi, PID control, encoder interfacing, data visualization

AVR Autonomous Line-Following Car with Smart Parking (Arduino Uno) *2026*

- Programmed an autonomous line-following robot using AVR (AT mega) in Microchip Studio with 3 IR sensors for line tracking and crossline detection logic
- Implemented a parking bay detection system using an IR receiver — upon single crossline detection, the car enters parking search mode, detects the bay, and parks autonomously
- On double crossline detection, the car exits parking mode and resumes normal line following, demonstrating a full FSM-based behavioral control system
- Stack: AVR (AT mega), Arduino Uno, Microchip Studio, Embedded C, IR sensors

3D CAD Design of a Fully Functional Robotic Arm (6 DOF) – Fusion 360 *2026*

- Stack: AVR (AT mega), Arduino Uno, Microchip Studio, Embedded C, IR sensors
- Designed a 6-DOF robotic arm from scratch in Fusion 360, modeling all links and joint assemblies
- Configured rotational joints and motion constraints using Fusion 360's joint tools to simulate realistic articulation
- Exported the design for ROS integration, [add specific detail here — e.g. "generating a URDF for use in Gazebo simulation
- Stack: Fusion 360, Mechanical CAD Design, ROS (URDF export)

ESP32-CAM Remote-Controlled Car with Live Video Streaming *2025*

- Built a WiFi-controlled RC car with live video streaming using the ESP32-CAM module as both the access point and video server
- Developed a mobile app and a web-based controller (live on Vercel) allowing real-time driving and camera feed viewing from any device on the network
- Integrated live MJPEG video stream with directional controls, synchronizing car movement and video feed over ESP32's internal WiFi
- Stack: ESP32-CAM, Embedded C, JavaScript, Vercel, WiFi/HTTP streaming

IoT Smart Surveillance & Access Control System (ESP32) *2024*

- Built a smart surveillance system integrating camera module with Telegram bot for real-time intrusion detection and remote door lock/unlock via commands
- Implemented UART/I2C communication protocols and real-time IoT messaging for reliable remote control
- Stack: ESP32, C/Embedded C, Telegram Bot API, IoT communication

FSM-Based Sequence Detector *2024*

- Designed a pseudo-random sequence generator using XOR feedback logic and PISO shift register serialization
- Implemented a Finite State Machine (FSM) for pattern detection with synchronized clocking

Learning Management System (Spring Boot) *2024*

- Developed RESTful backend APIs for authentication and course management with scalable database integration

CERTIFICATIONS & LEARNING

- *Mastering ROS 2 for Robotics Programming Specialization* — Packt (Coursera) — **In Progress**
- Introduction to Embedded Systems Software and Development Environments — University of Colorado Boulder (Coursera)
- The Complete Full-Stack Web Developer Bootcamp — Dr. Angela Yu, Udemy

INTEREST AREAS

Embedded Systems | IoT | Robotics & Control Systems | Network Engineering | Telecommunications | Industrial Electronics | Software Engineering

REFERENCES

Prof. Anuradha Jayakody

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